COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
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ACTIVITY NO. 2 **Prelim Laboratory Exam Set D**

1. PROCEDURE OUTPUT

ALGORITHM

QUEUE - LINKED LIST IMPLEMENTATION

Step 1. Creating a class node
Step 2. Create a newNode creator, then allocate a space for the new node.
Step3. Store the new data on the newNode then point to null
Step4. Check if the queue is empty, if empty, store the data into front.
Step4. Enqueue the next data into the front, and back pointing to the previous enqueued data.
Step5. Display all enqueued datas.
Step6. Check if datas are displayed symmetrically.
Step7. End

This algorithm follows the concept of Queue which is the FIFO (First in, First Out). Data which are stored first will be the first to be removed. Data is also read from top to bottom. In the linked list, once an item is inserted into an empty queue, the data stored in newPtr will become the front and back. The newPtr will be pointed to a new node or the next data, which will become the new back of the queue.

ALGORITHM

STACKS - ARRAY

Step1. Start
Step2. Define max capacity for stack - array.
Step3. Create Stack 1, then use push to set data into the top of the stack.
Step4. Read the stack from bottom to top.
Step5. Create Stack2. Stack 2 stores popped datas from Stack 1.
Step6. Use pop to remove the last pushed data.
Step7. The first data for Stack 2 will be the last data inserted from Stack 1.
Step8. See if the datas matches symmetrically.
Step9. END

This algorithm follows the concept of Stack which is the LIFO (Last in, First Out). In arrays, the maximum size of each program is fixed, meaning the program is set to their maxCapacity. The data on the stack will be moved, once a new data will be inserted making the first inserted data as the back. Once popped, the data will be popped from top to bottom, which means the last data to be entered will be the first one to be removed.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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